DIVYANSHU GOYAL

Multimodal Model Training | RL Fine-Tuning | Evaluation

Sunnyvale, CA | +1 404-414-3762 | divyanshu.g25@gmail.com

linkedin.com/in/divyanshu2594 | github.com/divyanshu25 | Google Scholar

SUMMARY

Applied scientist with 8+ years at Adobe, passionate about multimodal large language models. I lead model development end to end: data strategy, SFT and RL fine-tuning, reward design, evaluation, and serving. As tech lead in a 20+-person team, I took Adobe GenStudio for Performance Marketing from first prototype to a multi-million-dollar product used by Fortune 500 companies. I am now tech lead of a 7-person team building marketing VLMs, judge models, and photo-retouching models for subjective domains, where quality depends on human taste. With Adobe Research, I co-led ATune, a 54M-parameter retoucher whose edits human raters prefer to GPT-6 Astra's. My work rests on rigorous evaluation: leak-checked benchmarks, human preference studies, and paired statistical tests. I started my career building Adobe's high-throughput ML serving platform. I publish at CVPR, ICML, and AAAI workshops, hold patents in generative AI, and write about AI on my blog. I am a strong believer in first-principles thinking.

EDUCATION

Institution	Degree	GPA	Years
Georgia Institute of Technology	M.S. Computer Science (Machine Learning)	4.0/4.0	2019 – 2021
BITS Pilani	B.E. Computer Science, M.Sc. Mathematics	8.75/10	2011 – 2016

EXPERIENCE

Applied Scientist – Adobe Inc., San Jose, CA — June 2021 – Present

FixIt: Multimodal Models for Marketing and Photo Quality (Tech Lead, team of 7)

- Tech lead for a 7-person team, end to end across three models: a Gemma-3-12B GRPO policy, Qwen3.5-4B judges, and the ATune retoucher. Owned the annotation strategy, training pipeline, and evaluation protocol.
- Delivered sub-second multimodal inference for marketing decisions. Demos drove org-wide investment in this model class; customer engagements opened new revenue streams.

Adobe Marketing VLM (Gemma-3-12B, SFT and GRPO)

- Trained Gemma-3-12B (LoRA $r{=}256$, 1.19B trainable parameters) with distributed **SFT**, then multi-round **GRPO** on 8× H100: group-sampled completions per prompt, programmatic rewards against ground truth, clipped policy-gradient updates.
- Designed rewards for 7 tasks (marketing quality, distortion type/severity, emotion, audience, attention, creative attributes, photo observables). Diagnosed **reward hacking** through mode-collapse audits of rollouts; redesigned the scorers.
- Built an async VLM annotation pipeline with chunked checkpointing and crash recovery, plus a router–worker inference API across 4 GPUs.

JEV-Style Marketing Judge Models

- Built a 1.1M-example training corpus from 14 human-rated sources, with rater-disagreement distributions as soft labels. Trained **Qwen3.5-4B** LoRA judges on sampled data mixtures.
- Calibrated the score head on crowd ratings, lifting validation SRCC on KonIQ from 0.421 (reference) to **0.663** (paired 95% CI > 0). Built the 67K-item held-out **benchmark suite**, leak-checked with SSCD copy detection against all training images.
- Shipped releases v9–v19 behind a non-inferiority gate on that suite; served at 21 replicas across 8 H100s. As a pairwise judge for crop selection: **0.82** accuracy on GAIC vs. 0.71 for the score head.
- Detected vision-encoder collapse within 200 steps (image-blind checkpoints); added a vision-health guard to the training loop.

ATune: Unsupervised Photo Retouching (Co-Lead with Adobe Research)

- Co-led with Adobe Research (co-first author, under review); mentored an intern. Owned the synthetic data pipeline, training recipe, and evaluation protocol.
- **54M-parameter** model trained on 10M unedited photographs with no expert edits. Predicts slider settings for off-the-shelf editors (RawTherapee, darktable) through per-editor solvers.
- Human 2AFC study (157 raters, 3,524 judgments): beats GPT-6 Astra on the same editor (**54.3%** of decisive votes, 95% CI 50.3–58.3); ties Claude Opus 5 (50.5%, CI 46.8–54.2). On FiveK, PPR10K, and SPAQ, at least **8 of 9** published IQA judges prefer it to GPT-6 Astra.
- Data-scaling study (1M–10M photographs): each larger corpus wins or ties; no judge prefers a smaller one on any test set.

Adobe GenStudio for Performance Marketing (Tech Lead, team of 20+)

- Tech lead from inception to first prototype, then for the production API server integrating the UI with cloud LLM providers at scale. Launched October 2024; now a **multi-million-dollar** product used by Fortune 500 companies.
- Defined the evaluation framework for brand compliance of generated content.
- Fine-tuned custom models exclusively on a Fortune 500 enterprise customer's data, leading to a GenStudio custom-models deal.
- Built a multimodal **Photoshop agent** with vision-grounded tool use.

Evaluation Research

- **DistortBench** (first author; CVPR 2026 and ICML 2026 Workshops): 13.5K questions over 27 distortion types × 5 severity levels. Best of 18 VLMs (Qwen3.5-27B) scores **61.9%** (chance 25%): above the mean annotator (60.2%), below a 3-annotator majority vote (65.7%).
- **JudgeBreak** (co-author; ICML 2026 Workshop): linear probes on Qwen3-32B hidden states reach **78.1%** pairwise accuracy across 95 subjective settings vs. 70.1% for the best reward model or LLM judge.

Machine Learning Engineer – Adobe Inc., Bangalore, India July 2016 – August 2019

- Early engineer on Adobe's ML serving platform, which hosts internal models at scale, including field-aware factorization machines (FFM). Contributed to day-to-day planning of the end-to-end product.
- Owned serving throughput: 3,500 QPS/node at 0.3–0.6 ms p99 (Adobe Tech Blog) through a GC-free asynchronous event pipeline on LMAX Disruptor that eliminated back-pressure at peak load.
- Built the model lifecycle platform: versioned deployment, monitoring, staged rollout across environments.

PUBLICATIONS

DistortBench: Benchmarking Vision Language Models on Image Distortion Identification
D. Goyal, A. Eppa, V. Bannihatti Kumar | CVPR 2026 and ICML 2026 Workshops | arXiv:2604.19966

JudgeBreak: Is Taste Hidden in the Layers? Stress-Testing Reward Models and LLM Judges with Representation-Based Subjective Evaluation
V. Bannihatti Kumar, A. Narang, M. Anand, A. Eppa, **D. Goyal** | ICML 2026 Workshop | OpenReview

Curiosity-Driven LLM-as-a-judge for Personalized Creative Judgment
V. Bannihatti Kumar, **D. Goyal**, A. Eppa, N. Bhandari | AAAI 2026 Workshop, Oral | arXiv:2510.05135

PROJECTS

VibeNanoChat: GPT-2 Scale LLM Pretraining from Scratch 2026 | code
Decoder-only transformer (RoPE, RMSNorm, sliding-window attention) with single-parameter depth scaling and a custom distributed **Muon** optimizer (ZeRO-2 state sharding). Trained in BF16 with FlashAttention-3 at over 1M tokens/s on 8× H100. Fit isoFLOP curves over 40+ runs (44M–522M parameters) to locate compute-optimal model sizes. Write-ups: Going Fast and A Lazy Engineer's Guide to Scaling Transformers.

BrownBearAI: On-Device Photo Understanding for iOS 2026 | App Store
On-device face clustering and duplicate detection for iPhone, with a per-build regression gate: **99.4%** BCubed F1 clustering LFW identities among 1,000 distractors; 98.0% face-detection precision on WIDER FACE.

PATENTS

Title	Number	Year
Guiding Language Translation with Translation Documents Using ML	US20260050753	2026
Brand-Aligned Marketing Content Generation	US20250117829	2025
Cultural Adaptation of Images Using Generative AI	US20260051084	2026
Parameter-Efficient Fine-Tuned Models	US20250217193	2025
Video-to-Video Culturalization Using Generative AI	Approved for filing	2026
Learning to Be Surprised: A Curiosity-Driven Framework for Creative Judgment	Approved for filing	2026

TECHNICAL SKILLS

Research: VLM Training & Evaluation, LLM Pretraining & Scaling Laws, Benchmark Design, Human Preference Studies, LLM-as-a-Judge & Reward Models, Reinforcement Learning (GRPO), Reward Design, Linear Probing, Bootstrap Statistics

Tools: Python, PyTorch, Distributed Training (DDP, ZeRO-2), torch.compile, FlashAttention-3, Liger Kernels, Muon, Mixed Precision (BF16), LoRA, Hugging Face Transformers/PEFT, Weights & Biases